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Wet Soil For A Burning World

Designing a Transition Toward Regenerative Boreal Forest Fire Management Through Material-Led Design

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Abstract: Wildfires in boreal forest ecosystems are intensifying as climate change and extractive forestry practices produce drier, more vulnerable landscapes. Current responses centred on fire suppression and fuel reduction treat fire as an isolated event and not as a symptom of a deeper ecological imbalance. The project positions soil moisture retention as a critical leverage point shaping fire behavior, asking how design can contribute to a transition from an extractive to a regenerative regime in forest management that aligns with the forest cycles of regeneration and renewal. Developed in collaboration with PaperShell, a Swedish material company converting paper waste streams into certified biochar, the project connects speculative thinking within industrial networks. Two design interventions are proposed: Percolare, a ceramic-biochar piece that acts on the topsoil, slowly releasing carbon; and Carbon Pod Cycle, a biochar module system starting in forest nurseries and influencing roots to slowly build conditions for regeneration.

Keywords: Transition Design; sustainability transitions; material-led design; soil moisture retention; biochar; wildfire resilience; boreal forests

1. Introduction - The Problem Beneath the Fire

Boreal forests, vast ecosystems of conifer trees, store vast amounts of carbon beneath the soil, making them vital for global climate regulation and wildlife. But today, these ecosystems are among the most fire-affected on Earth. Wildfires in boreal forests are becoming more frequent and severe as climate change and intensive forestry practices create increasingly dry and vulnerable landscapes. Carbon emissions from forest fires increased by approximately 60% globally between 2001 and 2023, while emissions from boreal forests in Eurasia and North America nearly tripled over the same period, driven by warmer and drier climatic conditions (NASA, n.d.). In Swedish boreal forests, conditions such as heat, wind, low

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humidity, dense monocultures, and resinous conifer species like pine and spruce increase fire intensity, as demonstrated during the extreme wildfire season of 2018 (MSB, 2018). Due to climate change and a reinforcing wildfire feedback loop, wildfires are projected to increase in frequency and intensity by 50% by 2100 (United Nations Environment Programme, 2022).

Current forest management practices often operate within a short-term logic, where interventions focus on reducing immediate fire risk through suppression, clearing, and extraction. While these actions respond to present symptoms, they can reinforce longer-term vulnerabilities by removing organic material, reducing soil moisture, and disrupting ecological processes that support regeneration. This creates a reinforcing feedback loop: as forests become drier and less biodiverse, their capacity to regulate fire decreases, leading to more severe wildfires. Over time, this cycle weakens the forest system and diminishes its ability to sustain fire as a regenerative ecological process. Breaking this loop requires a shift from short-term control toward long-term ecological thinking, restoring carbon cycles and soil conditions as part of the forest's capacity to regulate itself.

This research positions soil moisture as a deep leverage point shaping fire behavior, biodiversity, and regeneration dynamics. When moisture levels decline, decomposition slows, fuel accumulates, and regenerative capacity collapses. This raises a central design question: how can design support soil moisture retention as a leverage point for strengthening boreal forests in both wildfire prevention and recovery?

This project explores how material-driven design, approached systemically, can support soil moisture retention as a strategy for strengthening boreal forest ecosystems through extended time frames. The project draws on Bruno Latour's understanding of Earth as a network of interdependent actors[1], approaching forests as active landscapes and living systems composed of humans and more-than-human relations. The intent is not to propose definitive solutions; instead, the work aims to make visible the slow processes and hidden layers that shape wildfire behaviour, encouraging reflection on current forestry practices, the values embedded in them, and the ecological dynamics they often overlook.

2. Background and Framework - From Extraction to Regeneration

[1] Bruno Latour proposes Gaia as a system that is neither purely natural nor external to human action, but one that reacts and responds to the transformations taking place within it. He argues that climate and environmental conditions are not fixed, but are continuously produced through interactions between atmospheric processes, living organisms, and human activity. This perspective informs the project's approach to soil, moisture, and carbon as active participants within the forest system. See Latour, B. (2017), Facing Gaia: Eight lectures on the new climatic regime

2.1 Designing for sustainability transitions

The project is grounded in Transition Design and sustainability transitions theory, emphasizing extended timeframes, interdependence, and systemic change when addressing complex socio-ecological challenges (Irwin & Kossoff, 2015; Goss et al., 2025). Within this context, the transition can be understood as a shift in forest management practices in relation to wildfires, from reactive, extraction-oriented responses such as salvage logging, toward regenerative approaches that recognize fire as part of a broader ecological cycle. Conventional post-fire practices often prioritize the rapid removal of burned material, disrupting soil systems and eliminating critical resources such as charred wood, which plays an important role in moisture retention, habitat formation, and nutrient stabilization.

This transition calls for rethinking post-wildfire landscapes as active systems of regeneration, where fire-derived materials contribute to long-term ecological recovery.

Within this transition, soil moisture retention is identified as a deep leverage point shaping fire behavior, biodiversity, and regeneration dynamics. By focusing on this leverage point, the project contributes to a broader shift away from extraction-driven, short-term forest management toward regenerative approaches aligned with ecological rhythms and other-than-human agency.

As Donella Meadows argues, “the least obvious part of the system, its function or purpose, is often the most powerful place to intervene” (Meadows, 2009). This perspective informs the project’s focus on below-ground processes that are often invisible in forest management and design practices. These practices prioritize short-term recovery and resource extraction, often disrupting long-term ecological processes such as moisture retention and habitat formation.

2.2 Fire, forestry and forest regeneration

In Swedish boreal forests, current forestry practices, characterized by monocultures, dense plantations, and the prevalence of resinous conifer species, have increased landscape vulnerability to wildfires. These conditions, combined with rising temperatures, drought, wind, and low humidity, contribute to faster fire spread and higher fire intensity. In response, controlled burning is often used as a management strategy to reduce fuel loads and support ecological processes; however, when absent, forests can become dense, nutrient-poor, and more susceptible to large-scale, uncontrolled fires.

Following wildfire events, post-fire interventions frequently prioritize rapid extraction and site “cleanup,” most commonly through salvage logging. Burnt wood and residual biomass, such as branches, needles, roots, and small-diameter logs, are often removed, piled, and either burned or processed as low-value biomass. While biomass energy is commonly framed as a renewable resource, this perspective often overlooks the ecological trade-offs involved, as well as the economic inefficiencies that frequently render such material underutilized. The

extracted biomass is frequently of low commercial value, raising questions about whether rapid removal outweighs the ecological functions that burnt wood would otherwise fulfil. Salvage logging therefore becomes a pivotal moment in shaping how fire-affected landscapes regenerate and how carbon continues to participate in the forest system.

From an ecological perspective, post-fire landscapes act as critical phases within forest regeneration cycles as opposed to the general perception of fire. Fire transforms wood by reducing cellulose and lignin, leaving behind carbon-rich, charred material that is more resistant to microbial decay. This burnt wood plays a significant role in forest ecosystems: it supports fungi, saproxylic insects, and bird species, contributes to soil formation, and acts as a long-term carbon reservoir. When left in site, it helps stabilize nutrients and provides structure for emerging ecological communities.

In contrast, the removal of burnt material through salvage logging disrupts these processes, reducing biodiversity and limiting the forest's capacity for natural regeneration. Public perceptions, however, often frame burned forests as devastated landscapes, reinforcing interventionist approaches that prioritize visual recovery and economic return over ecological function.

For local communities, these landscapes also carry cultural and emotional significance, becoming markers of environmental change and climate vulnerability. At the same time, ecologists emphasize that regeneration is already underway: many plant species are adapted to fire, resprouting from root systems or germinating from seeds triggered by heat. While immediate post-fire conditions may include risks such as erosion, flooding, or hydrophobic soils, longer-term recovery processes can unfold over decades to centuries.

Within this context, burnt wood occupies multiple potential pathways. It may remain in the forest, supporting biodiversity and soil development; it may be extracted and processed into biomass or biochar; or it may be recontextualized as a material within design practices. These trajectories highlight the tension between extractive and regenerative approaches to forest management, and position carbon as a key material in shaping future forest ecologies.

2.3 Soil moisture

Soil moisture strongly shapes how boreal forests burn and regenerate. The organic layer, mosses, litter, and humus, becomes increasingly flammable when moisture drops below 12–15%, and at levels under 10% it can sustain deep, smouldering combustion that burns into the soil and is difficult to extinguish. When the moisture level is higher than 25–30%, on the other hand, ignition is limited and fires tend to stay low-intensity, affecting surface vegetation instead of deeper soil carbon.

Soil moisture not only determines flammability spread, but also controls biological activity. In dry conditions, microbial and fungal processes can slow by up to 70%, reducing decomposition and allowing undecomposed material to accumulate. This build-up increases

fuel loads and reinforces a cycle in which dry soils lead to more intense fires, which in turn further dry the landscape.

Because small shifts in soil moisture affect fire behavior, decomposition, and long-term resilience, it emerges as a critical leverage point within boreal forest ecosystems

2.4 Forest interdependencies

Boreal forests are a complex network shaped by interdependent relationships. Deadwood is a central component of this network. It holds moisture, provides structure, and supports the organisms that drive decomposition. Among these are saproxylic insects, which depend on dead and decaying wood during parts of their life cycle. Their activity breaks down organic matter, releases nutrients, and contributes to the gradual formation of soil. Yet these species are increasingly vulnerable: in Europe, 17.9% of saproxylic beetle species are classified as threatened (Nieto & Alexander, 2018), due largely to forestry practices and post-fire interventions that remove the materials they depend on.

These relationships illustrate that soil moisture, carbon, and regeneration are not isolated factors but interconnected reinforcing processes. Moist soils support microbial and insect activity; their decomposition improves soil structure and carbon retention; and these factors affect how the forest burns and regrows. If dead wood is removed or there is less moisture in the soil, then this cycle will break down, thus decreasing diversity and slowing the recovery process of the forest.

By foregrounding these interdependencies, the project positions design within a broader ecological context. It acknowledges that interventions aimed at soil moisture must also consider the organisms, materials, and temporalities that shape how moisture moves and is retained in forest systems. This perspective informs the design explorations that follow, which seek to work with the complex relationships that enable boreal forests to regenerate over time.

3. Methodology - Design as a Systemic Inquiry

Methodologically, the project combines desk-based research, site-based fieldwork, expert interviews, and material-led design inquiry, guided by the Multi-Level Perspective (MLP). This approach treats design as a form of systemic inquiry, integrating ecological knowledge, material experimentation, and situated observation.

3.1 Desk- based research

Desk-based research examined global wildfire dynamics, soil moisture conditions, and forestry practices, situating the project within broader landscape-level pressures associated with climate change and shifting fire regimes. Desk-based research was synthesized through a comparative table within the Multi-Level Perspective positioning, identifying key insights like moisture distribution across soil layers and forest topographies.

Table 1 Research Translation Framework: The table synthesizes key insights gathered throughout the research process across multiple scales and domains, illustrating how ecological observations, material experimentation, and forestry-related studies informed the conceptual and systemic development of the proposed interventions.

Source	**Domain**	**Scale**	**Key Insight**	**Design Translation**	**Relevance to System**
Biochar Soil study	Soil science	Micro	Biochar increases water retention and microbial activity in porous structure	Informs material configuration	Defines soil moisture as a leverage point
Wildfire behavior research	Ecology	Macro	Fire intensity is strongly linked to soil moisture levels	Frames intervention focused on soil management	Supports system argument
Swedish forestry practices	Forestry	Meso	Salvage logging reduces soil carbon and biodiversity	Positions the need for alternative interventions	Frames problem context
Saproxylic species study	Ecology	Micro	Deadwood supports fungi, insects, and biodiversity	Informs biological role of biochar	Supports biodiversity dimension
PaperShell material process	Material/ Industry	Meso	Biochar can be produced from waste streams at scale	Connects design to industrial production	Enables scalability
Nursery systems research	Forestry/ agriculture	Meso	Early-stage intervention increases	Defines design intervention	Enables temporal strategy

		survival and distribution efficiency	integration in nursery system

3.2 Place- based fieldwork

This systemic framing was extended through site-based research in fire-affected boreal forest landscapes in the Stockholm region, including Tyresta National Park and Huddinge Naturreservat. Field observations focused on natural regeneration patterns, soil conditions, and interdependency networks between human and other-than-human actors. Site observations allowed study of how forests naturally regenerate. Two areas where fires had happened 10 years ago were analyzed. In both areas patterns of regeneration were visible, highlighting the growth of young species like lodgepole pine, aspen and birch, as well as shrubs and grasses. Traces of bark and saproxylic beetles were also tracked. These observations helped to analyze beneficial conditions of forest regeneration and apply them in later ideas for design interventions.

3.3 Expert interviews

Semi-structured interviews were conducted with experts in wildfire management, soil and forest ecology, material development, and saproxylic insects. These interviews provided situated knowledge that informed both system understanding and design decision-making.

3.4 Material- based design research

Material-led experimentation and prototyping enabled burnt wood and biochar to be explored as active agents within processes of moisture retention, decomposition, and soil regeneration. Across these methods, the research emphasizes learning, reflexivity, and cross-scale understanding, aligning design practice with the complexity and extended temporalities characteristic of sustainability transitions.

The research focused on biochar's properties of high porosity and water holding capacity as key elements. Tests were carried out to investigate how the biochar could be shaped into stable yet porous material configurations. Different variations combining biochar with soil, red clay, carbohydrate-rich binders and cellulose-based fibres were tested in a two part mold, varying binder and biochar ratios to balance integrity, permeability, porosity and aesthetic qualities. Parallel biochar and red clay variations were tested through additive manufacturing. A series of water retention tests were conducted to observe how these materials absorb, retain and release water over time. These iterations functioned as a testing ground through which material behavior informed subsequent design decisions.



Figure 1 Biochar porosities and material configurations. Photograph documenting material experimentation exploring different biochar particle sizes, porosity levels, and clay-biochar compositions within the development of the interventions. Photo by author

4. Design for Transition - Two Propositions Across Time

4.1 Bringing carbon to the soil

Biochar is a carbon- rich porous material resulting from burning biomass (such as wood, agricultural residues or plant fibres) through high temperature pyrolysis in a low oxygen environment. Through this process, methanol, acetic acid, and other oxygenated volatile organic compounds are released while the remaining carbon structure becomes highly stable and porous. Because of this structure, the material can remain in soils for long periods of time while retaining a complex internal structure of microscopic pores. Due to this pores, biochar obtains properties that have been proved beneficial for soil systems since it can increase soil water retention, slow absorption and release of moisture, improvement in the aeration and structure of the soil, provides habitat for microorganisms and supports nutrient cycling. Biochar can also retain nutrients and reduce leaching, helping soils maintain fertility over time. Although it is an ancient material, biochar has gained increasing attention in the past years in fields like agriculture, forestry and climate mitigation mainly for its soil health improving capacities and carbon storage through extended time frames.

In current systems, biochar is typically applied as soil amendment directly mixed into agricultural soils or forestry substrates. It is normally distributed in bulk quantities across

large areas or incorporated into growing media in nurseries. At an industrial scale, these systems often rely on large- scale material distribution and do not necessarily consider how biochar might function as a designed material within ecological infrastructures. At a local scale, new systems have emerged where biochar is locally produced and installed into nurseries or farming systems showing potential for community involvement and circular cycles for production.

4.2 Design Objectives

The objective of this project is to explore how biochar can act as an ecological infrastructure that supports soil moisture retention, regeneration and biodiversity in boreal forest ecosystems. The design-led project delves into how carbon- rich materials can be reintroduced into forest cycles through design interventions that operate across both systemic and communicative scales.
To ensure this, two complementary design explorations are in development. The first intervention acts on the topsoil and explores a clay and biochar printed piece that interacts with humidity and rainfall, capturing moisture and gradually releasing carbon into the soil, while making these slow processes visible to human stakeholders. The second intervention acts in a systemic way, addressing the roots. It investigates a biochar- based module system designed to be integrated into nursery planters, enabling carbon- rich material to accompany trees from the earliest stages of growth to gradually integrate into forest soils in extended time frames. By intervening at the nursery stage, the project explores how carbon can be distributed incrementally across future forests through existing planting systems.

The design intervention focuses on three interconnected objectives:

- Supporting soil moisture retention as a leverage point for wildfire prevention and post-fire regeneration.
- Exploring distributed carbon integration through nursery systems as a strategy for preparing future forests.
- Communicating slow ecological processes through material artifacts that make soil regeneration visible to human stakeholders.

These objectives are explored through material configurations that respond to ecological conditions, spatial placement within nursery planters and forest environments, and the long-term interaction between biochar, soil systems, and biodiversity.

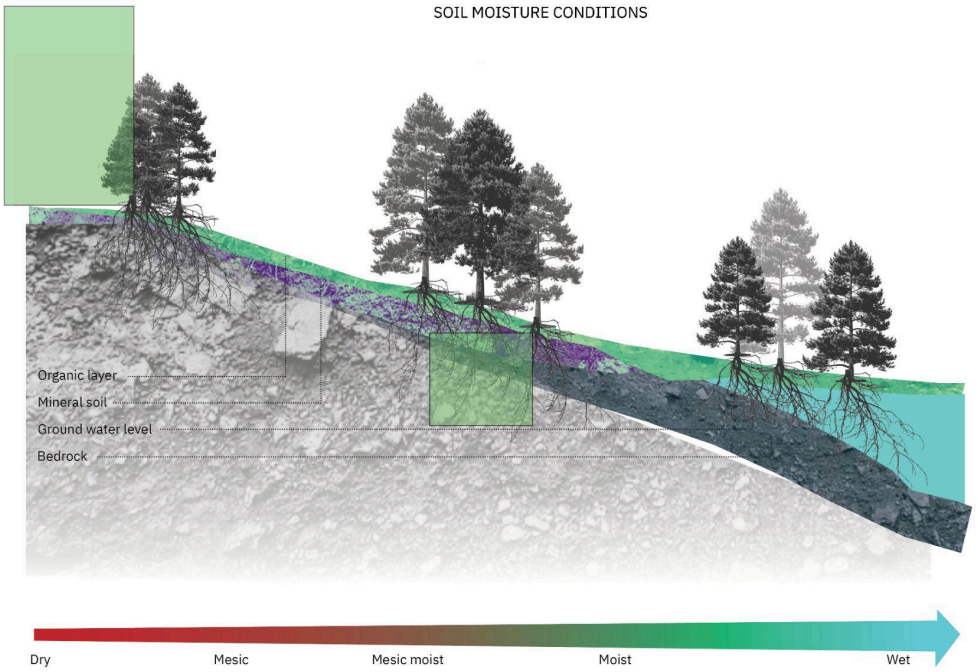


Figure 2 *Design interventions related to soil moisture conditions across a forest topographic gradient. The gradient ranges from dry upland conditions on the left to wet lowland conditions on the right, across four soil layers: organic layer, mineral soil, groundwater level, and bedrock. Higher and drier areas of the slope represent the conditions most vulnerable to fire ignition and fuel accumulation, and the primary spatial context for the design interventions.* *Illustration by author*

4.3 Design for Transition

As outlined earlier, this project explores how material-based design can support a transition from reactive wildfire management toward more regenerative forest systems operating across extended time frames. The project investigates how ecological conditions can gradually be reshaped through carbon distribution, soil regeneration, and long-term thinking.

Guided by principles of Transition Design, the project operates across both material and cultural dimensions. It does not propose a solution, but instead explores how design interventions can act as niche experiments within larger forest systems, engaging both ecological processes and the ways forests are perceived and managed.

Within current forest regimes, interventions often act at the surface level and respond to wildfire after disturbance occurs. In contrast, this project explores how regeneration might emerge through interventions operating across different temporal and spatial scales. As

illustrated in the transition diagram, the project positions design between material practices, systemic change, and shifts in posture and mindset.

The two proposed interventions therefore operate at different points within the system. One engages present forest conditions directly, acting as a visible and communicative point of entry into ecological processes that are often difficult to perceive. The other operates within future forest systems, embedding carbon gradually within root structures over extended timeframes.

Together, these interventions explore how design can contribute to a transition from short-term extraction and mitigation toward forest systems in which carbon, moisture, and biodiversity are gradually rebalanced over time. The project investigates how changing underlying soil conditions might support forests that are increasingly capable of sustaining regenerative processes on their own in relation to wildfires.

TRANSITION FRAMEWORK MAPPING

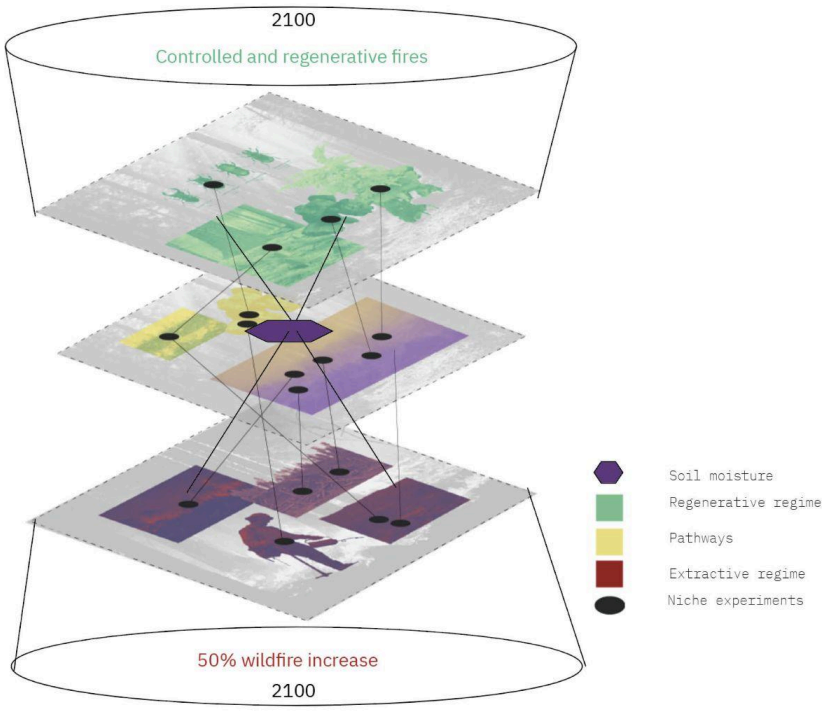


Figure 3 Transition framework mapping regenerative and extractive wildfire futures. Diagram illustrating the relationship between extractive forestry regimes, niche experiments, regenerative pathways, and soil moisture as a leverage point within long-term wildfire transition scenarios. The framework visualizes potential trajectories between increasing wildfire intensity and regenerative fire ecologies toward the year 2100. Illustration by author.

5. Design Intervention

5.1 Percolare

Acting on soil moisture is often hidden and not visible to human actors, even though it plays a key role in forest systems. Healthy soil supports up to 80% of forest biodiversity, and this needs to be communicated more clearly. The idea for *Percolare* comes from the desire to bring attention to this and to promote action around it. By showing how the material works and how it interacts with the soil, it can support a better understanding and implementation of biochar, and potentially lead to a wider use in forest production.

The structure itself is both functional and communicative, and is intended to be placed in forests, whether in natural reserves or managed landscapes. Each structure contains a large amount of biochar bound with red clay and fertilizer.

When placed in the forest, it captures water from rainfall, retains it through its porous material, and slowly releases carbon into the soil as it degrades. When positioned in higher and drier areas of the forest, it allows for a more equal distribution of moisture in the topsoil, and over time into deeper layers.

Each piece contains around 50 kg of biochar, and through its production and placement is estimated to sequester approximately 183 kg of CO_2.

To communicate the properties of biochar and how it acts on the soil, the formgiving draws on three main principles: capture, retain and slowly release. The structure follows an hourglass-like shape, following a ring-like spiral to maximize the internal area and therefore more sequestered carbon. The intervention is named Percolare, derived from the Italian word percolare, referring to the slow movement of water through porous layers of soil and organic matter.

The chosen fabrication method is large format additive manufacturing using a clay extrusion system with an 8 mm nozzle. This process allows for a more controlled extrusion of the biochar and clay material through different layers, and it enables the production of internal geometries that would be difficult to produce otherwise using conventional casting or molding techniques. The additive process also allows for the intentional design of internal

voids and surface patterns, which play a role in water absorption, retention, and gradual release as well as maximizing the amount of biochar that can be sequestered per structure.

When placed in the forest, the structure begins to integrate into the soil through its interaction with moisture and biological activity. Rainfall and atmospheric moisture are absorbed by the porous material, allowing the structure to retain moisture and slowly release it into the surrounding soil. This creates a localized increase in soil moisture, especially in the upper layers where conditions are often driest. Over time, the material starts to wear and crack due to repeated wetting and drying cycles, as well as microbial and fungal activity. As the structure gradually degrades, biochar particles are released and begin to mix with the surrounding soil, allowing carbon to move from the surface into lower layers, supported by water infiltration, root growth, and soil organisms.



Figure 4 "Percolare" fabricated through robotic clay-biochar extrusion. The layered surface and hourglass geometry maximize internal volume and porosity for moisture retention and gradual carbon release, as well as creating microhabitats for different species. Photo by author

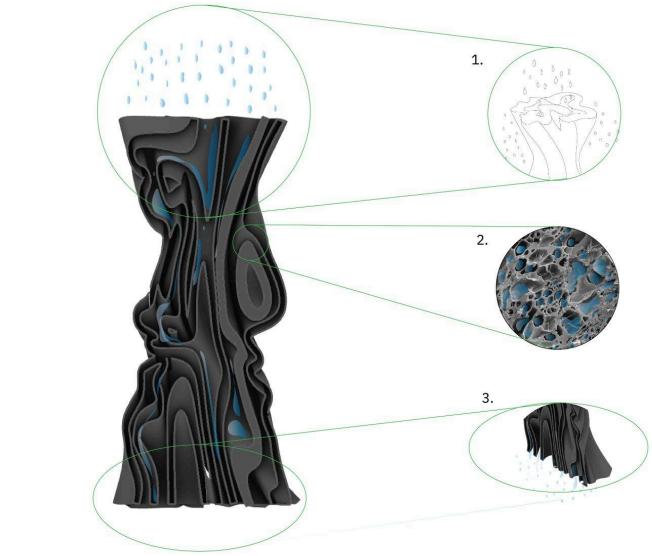


Figure 5 Water capture and moisture retention in "Percolare". Diagram illustrating how the porous clay-biochar structure captures rainfall and moisture, retains water, and gradually releases moisture and carbon through its internal cavities. Illustration by author

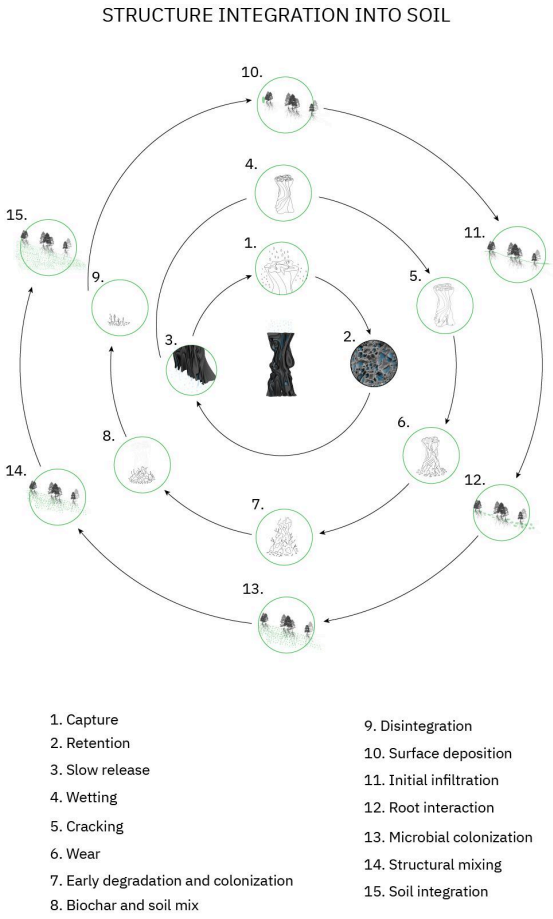


Figure 6 "Percolare" Integration into soil structures. System diagram illustrating the projected ecological integration of the Percolare intervention over time. The framework maps processes including water capture, retention, weathering, carbon cycling, microbial colonization, root infiltration, and soil integration, emphasizing the gradual participation of the intervention within regenerative forest ecosystems. Illustration by author

5.2 Carbon Pod Cycle

Current practices for introducing biochar into soil systems typically focus on either prevention or post-fire regeneration and often depend on large-scale distribution methods, such as aerial spreading by helicopter or transport by trucks to reach affected forest areas. While these approaches demonstrate the potential benefits of biochar, they rely on significant logistical infrastructure and tend to occur after ecological disruption has already taken place.

The project explores this question by proposing shaping the biochar material into compressed modules that act as structured elements within agroforestry systems. By integrating these pods into nursery planters, carbon-rich material can accompany trees from their earliest stages of growth and gradually enter forest soils as seedlings are planted and forests develop. In this way, the project investigates how biochar might operate not only as a soil additive but as a localized ecological infrastructure capable of supporting moisture retention, microbial activity, and long-term soil regeneration with immediate action in the roots and even distribution.

The material consists of a combination of clay, 80% biochar and molasses, since it is a carbohydrate-rich material that will attract saproxylic insects. Each pod contains around 80 g of pure biochar. The Papershell biochar used in this project contains approximately 89% total carbon on a dry basis, according to the laboratory analysis, meaning that each module contributes approximately 71 g of stable carbon to the soil system.

The form development is based on concepts such as moisturizing, nourishing, replenishing, ripples, and regeneration, together with the intrinsic properties of biochar. The shape takes inspiration from natural elements such as water drops and pebbles, as these already communicate qualities of softness, accumulation, and flow. This results in a form that is curved and fits within the palm of the hand, allowing it to be easily handled during placement.

Positioned within the nursery planter at approximately 5-10 cm below the soil surface, the module allows the material to act directly within the root zone from the beginning. Once the seedling is transferred into the forest, the module continues to interact with the surrounding soil. As roots grow and establish, the biochar begins to mix with the soil, supporting moisture retention, stabilizing nutrients, and creating microhabitats for soil organisms, including

saproxylic insects. Over longer periods of time, carbon is gradually distributed through different soil layers, contributing to improved soil structure and increased biological activity. In the long term, this leads to forest regeneration, where the accumulated effects of moisture retention, microbial networks, and carbon storage begin to influence the resilience of the forest.

Nurseries are often concentrated areas of seedlings where you can maximize the area of the module placement without needing transportation and distribution in later phases of forest development. In Sweden, companies like Essunga Plantskola and small farmers have started to include biochar in the nursery planters.

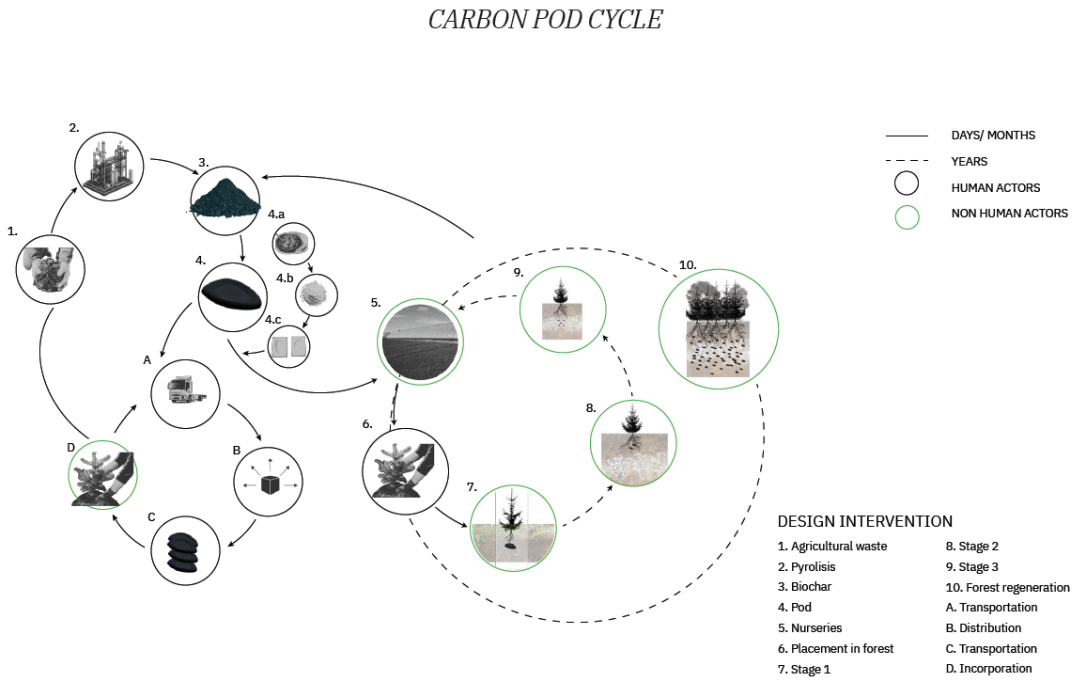


Figure 7 Carbon Pod Cycle System Diagram. Diagram illustrating the lifecycle and ecological integration of the Carbon Pod intervention, from agricultural waste and pyrolysis processes to biochar production, nursery integration, forest placement, and long-term regeneration stages. Illustration by author



*Figure 8 Pod placement and scale studies. Photographs illustrating the human scale and manual placement of the pod intervention within nursery and forest contexts.*Photo by author



Figure 9 Exhibition display of Carbon Pod Cycle, Konstfack Spring Exhibition 2026. The pods are presented alongside printed research documentation and loose biochar material, reflecting

the project's positioning between ecological research and material design practice. Photo by author

5.3 Collaboration with PaperShell

The project is currently in development through a collaboration with PaperShell, a Swedish materials company working with molded fiber and biochar within a circular production model. PaperShell converts production waste into stable biochar through pyrolysis and applies it to improve soil structure, moisture retention, and biodiversity, creating a closed-loop system that provides a grounded, real-world context for experimentation. Through this collaboration, the project gains access to biochar materials, nursery planters, and production knowledge, enabling controlled experimentation under consistent conditions.

The design intervention is conceived as a material trajectory that begins within industrial and nursery contexts and may later transition into forest environments, where biochar can participate in longer-term regeneration processes. The project aimed to work within industrial contexts, acknowledging both the opportunities and the limitations inherent in attempting ecological transition inside established systems. The project builds on an existing material stream but introduces a way for that material to be positioned, retained, and activated within forest soils, connecting industrial processes with long-term ecological regeneration.

5.4 Time Scale of Interventions

This proposal works across different time scales, positioning both interventions in relation to current forest conditions and future ecological processes. *Percolare* and *Carbon Pod Cycle* act at different moments and in different layers of the system, but both are oriented toward improving soil conditions over time. *Percolare* is introduced directly in the forest, where it starts working immediately at the surface, while the *Carbon Pod Cycle* is introduced earlier, within nursery systems, remaining stable and contained until it is placed in the soil together with the seedling.

In the short term, the structure supports immediate moisture retention at the surface, while the module supports early root development. Over time, both contribute to maintaining soil moisture and biological activity. In the longer term, they influence how carbon is stored across soil layers and how the forest is able to regenerate under changing conditions. Soil moisture plays a critical role in how fire spreads, how intense it becomes, and how the forest recovers afterwards. By acting both at the surface and within the soil, and across different time scales, the interventions contribute to creating conditions that can reduce fire intensity and support regeneration over time.

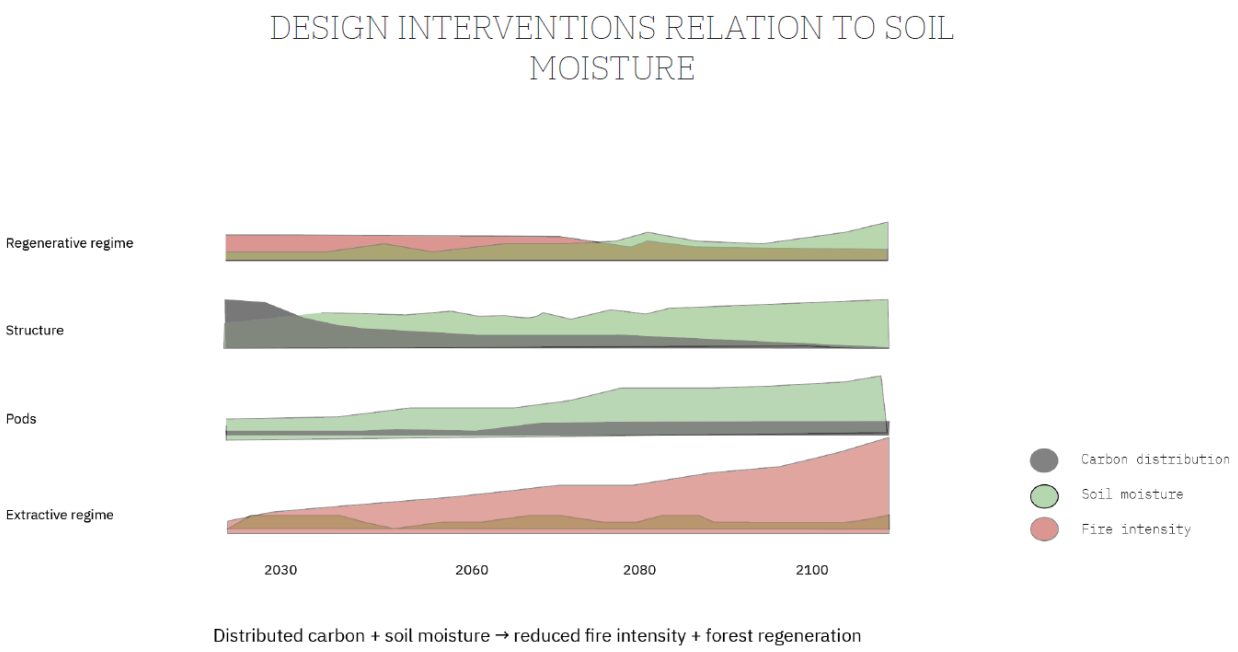


Figure 10 Diagram illustrating the projected relationship between carbon distribution, soil moisture levels, and wildfire intensity across regenerative and extractive forest regimes. Illustration by author

The diagram (See Fig. 10) visualizes the relationship between carbon distribution, soil moisture, and wildfire intensity across regenerative and extractive forest regimes over time. In the extractive regime, low levels of distributed carbon correspond with decreasing soil moisture and increasing wildfire intensity, reinforcing conditions of ecological vulnerability and degradation.

In contrast, the two proposed interventions operate together to gradually increase carbon distribution throughout the landscape and root systems, contributing to higher soil moisture retention over extended periods of time. *Percolare* acts at the landscape scale by capturing and slowly dispersing water and carbon through its porous structure, while the *Carbon Pod Cycl*e integrates carbon directly into root systems as forests grow and regenerate, through a series of cycles. Together, the interventions create overlapping layers of moisture retention and carbon integration that support regenerative ecological conditions. Within the diagram, increasing soil moisture is directly associated with reduced wildfire intensity and stronger regenerative forest dynamics, positioning soil moisture as a critical leverage point within long-term wildfire transitions.

6. Insights and Emerging Lessons

6.1 Soil Moisture as a Leverage Point

A central insight emerging from the project is that soil moisture functions as a deep leverage point shaping multiple ecological processes simultaneously, including fire behavior, decomposition, and biodiversity. When moisture levels decline, these processes slow or collapse, leading to fuel accumulation and reduced regenerative capacity. This underscores the importance of shifting design attention away from visible fire events toward below-ground and long-term system conditions.

Learning from natural regeneration processes revealed that burnt wood already performs essential ecological functions, such as moisture retention and carbon sequestration, but these are often disrupted by short-term management practices.

Material experimentation further demonstrated how design can operate as a form of systemic inquiry. Prototyping with biochar made slow ecological processes, such as decomposition and moisture regulation, more legible and discussable across disciplines and stakeholder groups. Together, these insights suggest that transition-oriented design benefits from engaging extended ecological timeframes and treating materials as active participants in regenerative systems.

6.2 Material experimentation as ecological study

Throughout the research, soil conditions, carbon distribution, and regeneration emerged as deeply interconnected ecological processes operating across extended temporal scales. Material experimentation became a way of engaging these systems not only conceptually, but directly through practice. Working with biochar began with ecological observation, understanding how burnt wood behaves in forest soils, and gradually shifted into technical investigation, exploring porosity, clay mixtures, degradation rates, and fabrication methods such as additive manufacturing.

Prototyping became a way of testing relationships: how moisture moves through porous structures, how carbon integrates into soil layers, and how microbial and fungal activity interacts with material forms over time. The process therefore shifted from producing isolated artefacts toward investigating how materials participate in the slower, less visible dynamics of soil formation and regeneration. This approach suggests that material-driven design can operate as a mediating practice, one that makes hidden dependencies, temporalities, and interactions within forest systems more legible.

6.3 Human and more- than-human relationships

The project engages multiple socio-ecological scales by placing material experimentation within the existing systems that shape boreal forests. These systems involve not only human actors: forest managers, nurseries, industrial partners, and researchers; but also more-than-human participants such as soil organisms, fungi, insects, moisture gradients, and the trees themselves. Their interactions collectively influence how forests regenerate over time.

The research exposed the limits of attempting to engage more-than-human systems through design. While the project seeks to foreground soil processes, ecological interdependence, and the agency of materials, these dynamics are always interpreted and mediated through human frameworks of understanding. Design remains a human-led practice, shaped by cultural expectations, institutional structures, and representational tools. This produces an intrinsic tension: the work aims to make room for more-than-human agency, yet it cannot escape acting from within human-oriented systems of meaning and intervention.

The research suggests that design may contribute more thoughtfully by acknowledging uncertainty, by resisting the urge to fully control ecological processes, and by positioning itself as a mediator between human perception and the slow, less-visible dynamics of forest systems. This perspective shifts design toward a more attentive role, one that works with ecological processes by supporting conditions for regeneration and not trying to control them.

6.4 Temporal conflict and forestry systems

One of the clearest insights emerging through the collaboration with PaperShell was the complexity of introducing new approaches within existing forest management structures. Contemporary forestry systems are shaped by long planning horizons, investment cycles, and industrial processes that aim to reduce uncertainty. This made visible how regenerative approaches often do not conflict with the values of stakeholders, but with the temporal frameworks through which forestry is organized.

The project therefore encountered a temporal tension between ecological time and industrial time, which shaped how interventions could be imagined, communicated, and potentially integrated. An unexpected finding was the extent to which communication and visibility became central to the design interventions themselves. What began as material prototypes gradually evolved into tools for engagement, making the role of biochar and soil processes more legible to different stakeholders. Percolare, in particular, functioned not only as an ecological proposal but also as a device that could prompt conversations around soil conditions, regeneration, and long-term forest strategies.

Feedback received throughout the research highlighted the importance of linking ecological considerations with the practical and economic realities that guide forest management. This suggests that transitions within forestry may rely as much on communication and shared understanding as on new practices or materials.

6.5 From extractive to regenerative loops

Within this proposed shift, the design artefacts function less as solutions and more as prompts for rethinking the cycles that shape forest management. By making soil processes visible and by engaging with biochar as a long-term material, the interventions invite a move away from the extractive loop driven by rapid suppression and short-term mitigation that gradually weakens forest systems. Instead, they point toward a regenerative loop in which carbon is retained, moisture is supported, and forest resilience increases over time.

In this framing, the artefacts encourage new ways of understanding how small, distributed, and time-based actions can participate in strengthening forests. They show how long-term thinking and material engagement might contribute to conditions where fire intensity decreases, carbon remains in the system, and forests become more capable of regenerating across extended temporal scales.

7. Insights and Emerging Lessons

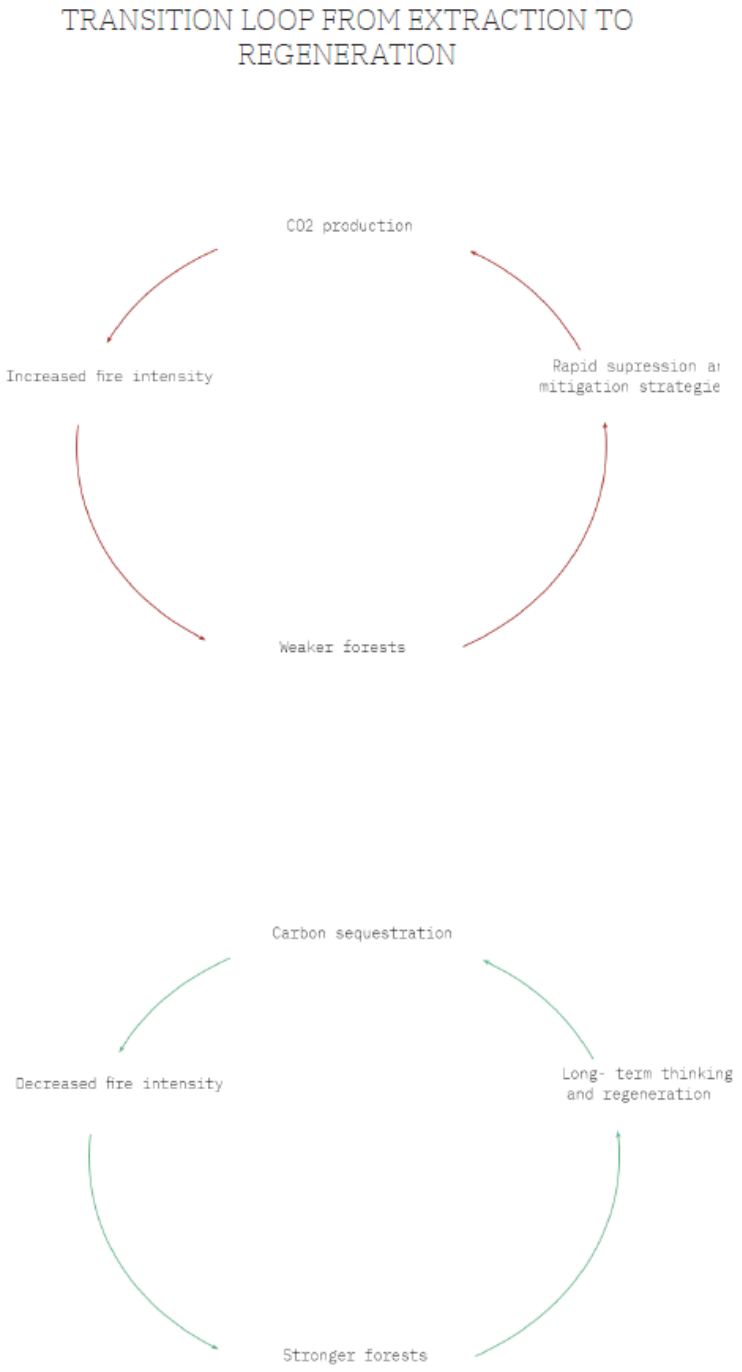


Figure 1 Transition loop from wildfire feeding cycle to wildfire decreasing cycle. Diagram illustrating the proposed transition from extractive wildfire systems that reinforce weaker forests and increasing fire intensity toward regenerative cycles that support carbon sequestration, stronger forests, and reduced fire intensity. *Illustration by author*

6.5 The exhibition as feedback

The project was exhibited as part of the Konstfack Spring Exhibition 2026 in Stockholm, Sweden, held from 13 to 20 May under the theme Connecting the Dots. Receiving around 13,000 visitors valuable information and feedback from visitors was gathered. Both *Percolare* and *Carbon Pod Cycle* were selected as the primary exhibited artefacts, framed as the material outcomes of the research while the surrounding space documented the process and research behind them.

A notable observation was that many visitors came from non-artistic backgrounds and found the research-oriented display more approachable and legible than the artistic presentation. This suggests that for this particular project, situated between ecological research and design practice, a more integrated presentation of research and outcome may better serve diverse audiences.

Two talks were held during the exhibition period, combining a video screening with a deeper explanation of the project. Both sessions attracted a small number of researchers, forest managers and practitioners involved in biochar production, generating meaningful discussions around future implementation, the intended scope of the proposals and the existing evidence base for biochar applications in boreal forest contexts.

7. Limitations and future directions

This project remains exploratory, and several limitations became evident throughout the research. While material testing generated initial insights into porosity, moisture behaviour, and design artefacts, the long-term ecological performance of the interventions cannot yet be evaluated. Seasonal cycles, microbial activity, and changing soil conditions unfold over years, and understanding how these interventions behave across such temporal spans would require extended field studies.

Scalability and implementation also remain open. The project does not claim to offer ready-made solutions for forestry systems; it acknowledges the challenges of integrating small-scale speculative interventions into infrastructures shaped by risk management, efficiency, and long-term planning.

A related limitation concerns how regeneration itself is assessed. Existing forestry metrics often focus on visible indicators, such as seedling establishment, growth rates, or canopy closure, while the slower, below-ground processes that support resilience remain largely unmeasured. Soil moisture, microbial activity, deadwood-dependent species, and carbon integration operate across temporal scales that do not fit easily within standard monitoring frameworks. This raises the question of what successful regeneration means, and whether

evaluation tools must expand to better reflect ecological function and not only short-term productivity.

The project also reveals conflicts embedded in the depiction of the ecological system through an artifact. While the intervention helps to illustrate soil processes which take time, translation from complex interactions to tangible forms necessarily involves reductionism, since interactions remain uncertain and unknowable.

Future work could investigate how regenerative strategies might align with existing forestry infrastructures and economic frameworks, as well as examine long-term biochar integration within diverse soil conditions and forest types.

7. Conclusion

As wildfires intensify under climate change, this project argues that forest resilience depends less on reactive control strategies and more on reconfiguring the ecological conditions that shape fire behavior over time. By positioning soil moisture as a deep leverage point and biochar as a material agent within long-term regeneration processes, the work reframes wildfire prevention and recovery as interconnected systemic challenges.

Through material-led experimentation, place-based research, and cross-scale collaboration, the project demonstrates how design can contribute to long-term sustainability transitions by intervening in feedback loops related to moisture retention, decomposition, biodiversity, and fuel accumulation. The research advances an experimental, transition-oriented design hypothesis grounded in learning, adaptation, and interdependence.

The contribution of this research lies less in delivering a defined solution and more in demonstrating how design can engage with systems that unfold across extended timescales. By working with material experimentation and ecological inquiry, the project positions design as a practice capable of operating within uncertainty and interdependence while opening space for alternative approaches to forest regeneration.

Fire is not the problem. The conditions that make it uncontrollable are.

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